

HW2 – Ying Zheng

Problem 1 (a)

```
library(Ecdat)

## Loading required package: Ecfun
## Warning: package 'Ecfun' was built under R version 3.5.2
##
## Attaching package: 'Ecfun'
## The following object is masked from 'package:base':
##
##      sign
##
## Attaching package: 'Ecdat'
## The following object is masked from 'package:datasets':
##
##      Orange
data(CRSPday)
crsp=CRSPday[,7]
crsp = as.numeric(crsp)
arima(crsp,order=c(1,0,0))

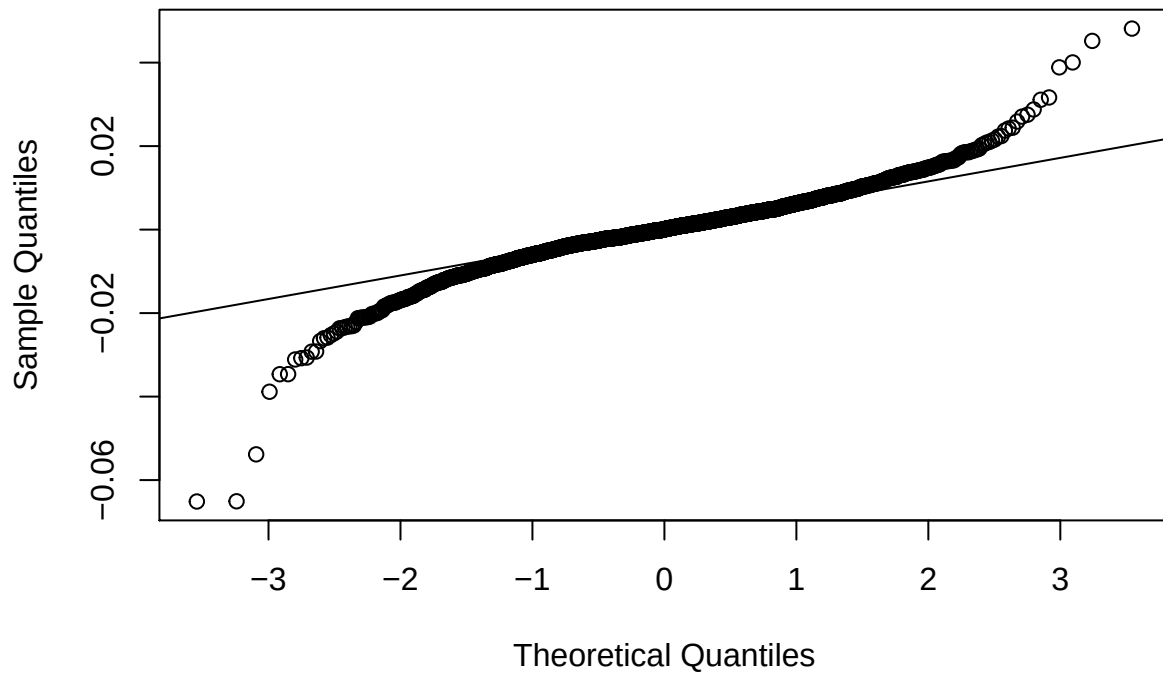
##
## Call:
## arima(x = crsp, order = c(1, 0, 0))
##
## Coefficients:
##          ar1  intercept
##          0.0853      7e-04
## s.e.  0.0198      2e-04
##
## sigma^2 estimated as 5.973e-05:  log likelihood = 8706.18,  aic = -17406.37
arima(crsp,order=c(2,0,0))

##
## Call:
## arima(x = crsp, order = c(2, 0, 0))
##
## Coefficients:
##          ar1          ar2  intercept
##          0.0865   -0.0141      7e-04
## s.e.  0.0199    0.0199      2e-04
##
## sigma^2 estimated as 5.972e-05:  log likelihood = 8706.43,  aic = -17404.87

a) Normal probability plot of the residuals: both residuals of AR(1) and AR(2) shows strong nonlinearity in
Q-Q plots (deviate a lot from the indicated straight line), which suggests correlations among residuals.
AR1=arima(crsp,order=c(1,0,0))
AR2=arima(crsp,order=c(2,0,0))
```

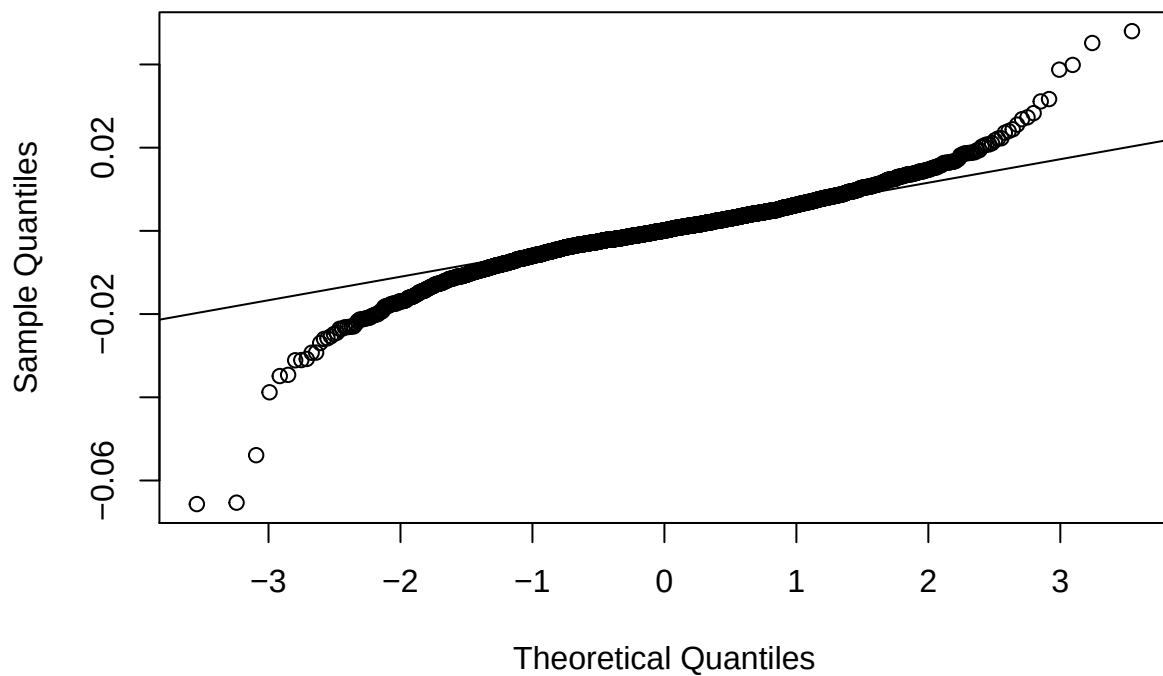
```
qqnorm(AR1$residuals)
qqline(AR1$residuals)
```

Normal Q-Q Plot



```
qqnorm(AR2$residuals)
qqline(AR2$residuals)
```

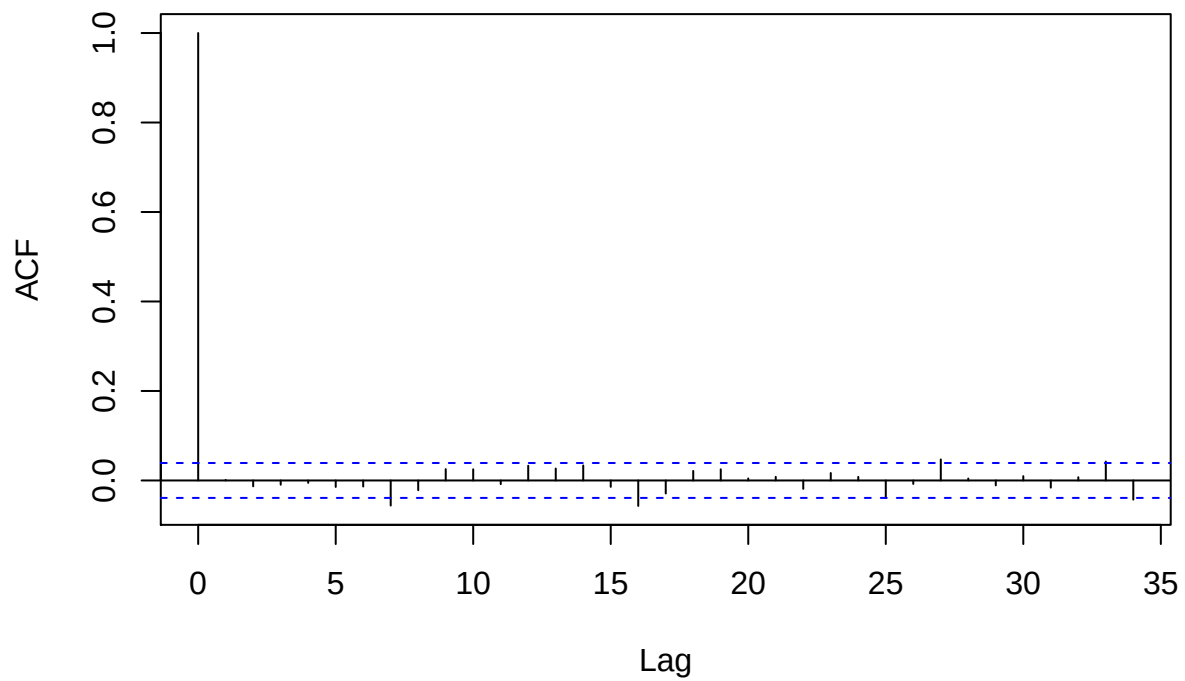
Normal Q-Q Plot



- b) ACF of the residuals: correlations in residuals for further lags still exist, indicated by significant acf of AR(1) with lag greater than 1 and significant acf of AR(2) with lag greater than 2.

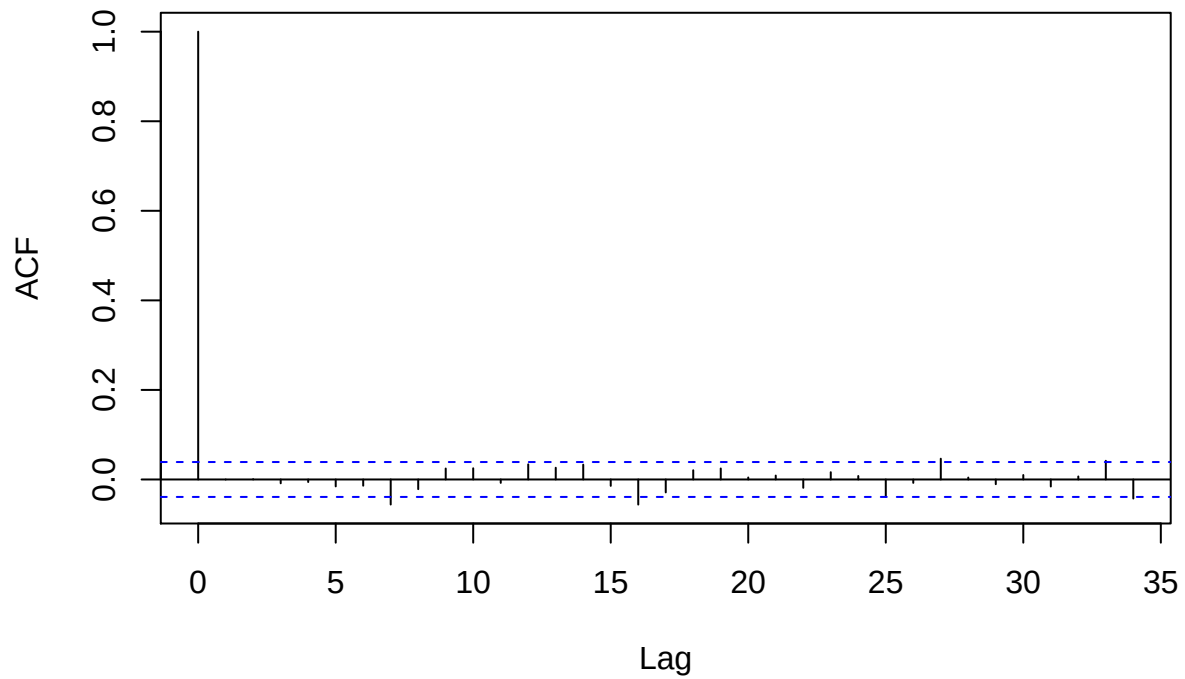
```
AR1=arima(crsp,order=c(1,0,0))  
AR2=arima(crsp,order=c(2,0,0))  
acf(AR1$residuals)
```

Series AR1\$residuals



```
acf(AR2$residuals)
```

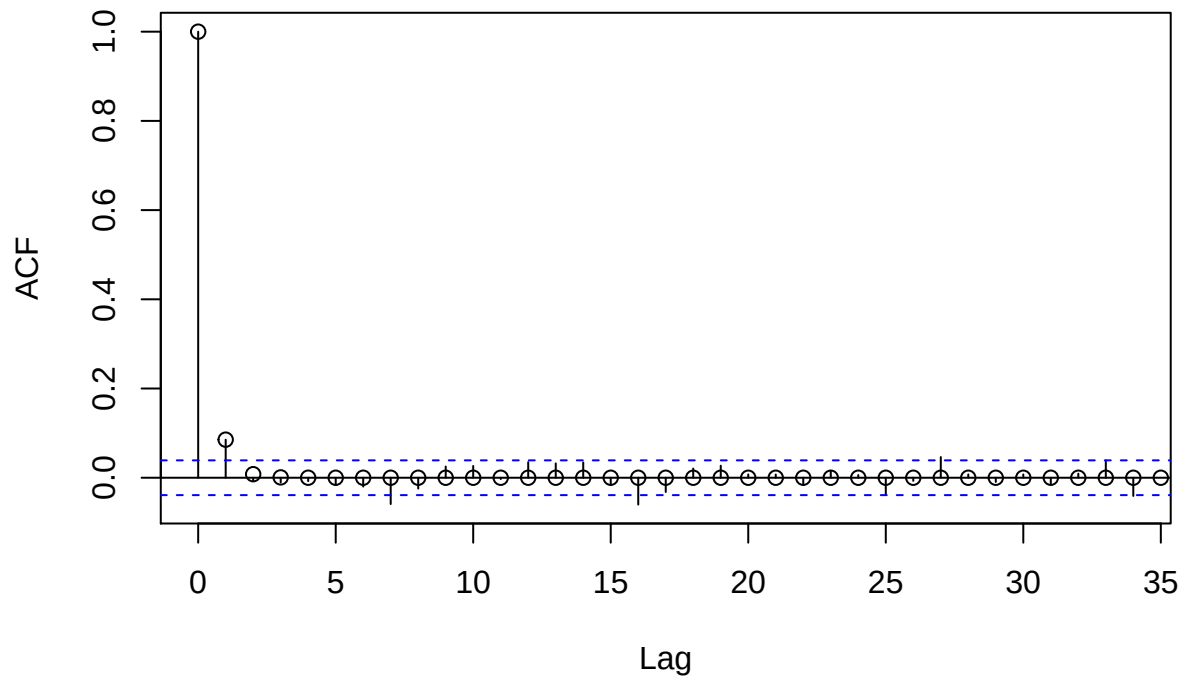
Series AR2\$residuals



c) Sample ACF for the data VS. the fit ACF for the model: sample ACF and fitted ACF are almost the same.

```
#sample acf
acf(crsp)
fit.acf1=ARMAacf(ar=AR1$coef,lag.max=35)
points(0:35,fit.acf1)
```

Series crsp

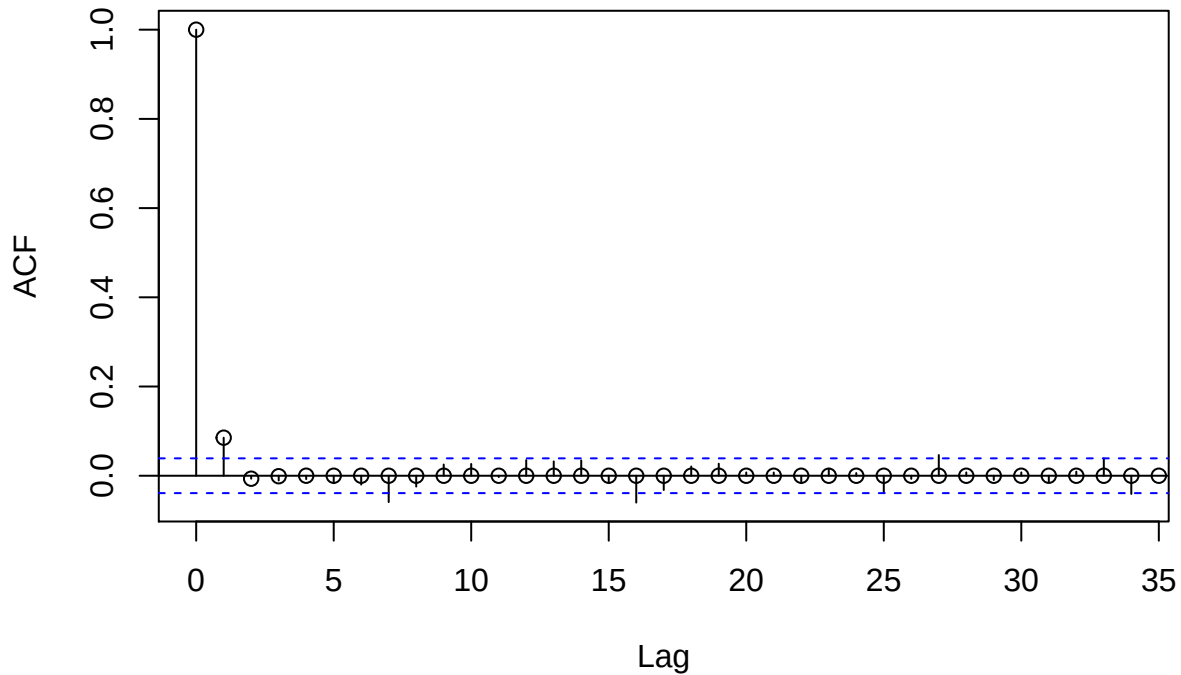


```
fit.acf1
```

```
##          0          1          2          3          4
## 1.000000e+00 8.535976e-02 7.960165e-03 7.369607e-04 6.826763e-05
##          5          6          7          8          9
## 6.323618e-06 5.857577e-07 5.425881e-08 5.026001e-09 4.655591e-10
##         10         11         12         13         14
## 4.312480e-11 3.994656e-12 3.700255e-13 3.427551e-14 3.174945e-15
##         15         16         17         18         19
## 2.940956e-16 2.724211e-17 2.523441e-18 2.337466e-19 2.165198e-20
##         20         21         22         23         24
## 2.005626e-21 1.857814e-22 1.720895e-23 1.594068e-24 1.476587e-25
##         25         26         27         28         29
## 1.367764e-26 1.266962e-27 1.173589e-28 1.087097e-29 1.006979e-30
##         30         31         32         33         34
## 9.327659e-32 8.640223e-33 8.003450e-34 7.413606e-35 6.867233e-36
##         35
## 6.361127e-37
```

```
acf(crsp)
fit.acf2=ARMAacf(ar=AR2$coef,lag.max=35)
points(0:35,fit.acf2)
```

Series crsp



```
fit.acf2
```

```
##          0          1          2          3          4
## 1.000000e+00 8.531014e-02 -6.667642e-03 -1.102014e-03 5.657583e-05
##          5          6          7          8          9
## 1.591776e-05 -1.683737e-07 -2.007353e-07 -4.195480e-09 2.354459e-09
##         10         11         12         13         14
## 1.267327e-10 -2.509391e-11 -2.361845e-12 2.356013e-13 3.668029e-14
##         15         16         17         18         19
## -1.751952e-15 -5.091990e-16 5.538072e-18 6.473791e-18 1.366011e-19
##         20         21         22         23         24
## -7.574693e-20 -4.089418e-21 8.073591e-22 7.616080e-23 -7.573371e-24
##         25         26         27         28         29
## -1.181975e-24 5.622852e-26 1.640135e-26 -1.758715e-28 -2.084412e-28
##         30         31         32         33         34
## -4.428410e-30 2.437923e-30 1.320129e-31 -2.597244e-32 -2.455727e-33
##         35
## 2.434531e-34
```

- d) Ljung-Box Test on the residuals: p-values are 0.01212 and 0.01083 respectively, which suggests strong evidence of correlation (h) not equal to 0 for some further h. Since p-value for AR(1) model is slightly larger than AR(2) model, AR(2) model is preferable in this case (more likely to reject null).

```
Box.test(AR1$residuals,lag=20,fitdf=1, type="Ljung-Box")
```

```
##
## Box-Ljung test
##
## data: AR1$residuals
## X-squared = 35.51, df = 19, p-value = 0.01212
```

```
Box.test(AR2$residuals,lag=20,fitdf=2,type="Ljung-Box")
```

```
##  
## Box-Ljung test  
##  
## data: AR2$residuals  
## X-squared = 34.53, df = 18, p-value = 0.01083
```

e) Value of AICc for each of the models: since AIC is slightly smaller for AR(1), we prefer AR(1) in this case.

```
print(AR1$aic)
```

```
## [1] -17406.37
```

```
print(AR2$aic)
```

```
## [1] -17404.87
```

Problem 1 (b)

```
# Find a 95 % confidence interval for phi for the AR(1) model.  
arima(crsp,order=c(1,0,0))
```

```
##  
## Call:  
## arima(x = crsp, order = c(1, 0, 0))  
##  
## Coefficients:  
##          ar1  intercept  
##       0.0853       7e-04  
## s.e.  0.0198       2e-04  
##  
## sigma^2 estimated as 5.973e-05: log likelihood = 8706.18, aic = -17406.37  
CI=c(0.0853-0.0198*1.96,0.0853+0.0198*1.96)  
CI
```

```
## [1] 0.046492 0.124108
```

Problem 2

```
#Remember to set max.q = 0  
library(forecast)
```

```
## Warning: package 'forecast' was built under R version 3.5.2
```

```
##  
## Attaching package: 'forecast'  
## The following object is masked from 'package:Ecfun':  
##  
##      BoxCox
```

```
auto.arima(crsp,max.q=0,approximation=F)
```

```
## Series: crsp  
## ARIMA(1,0,0) with non-zero mean  
##  
## Coefficients:  
##          ar1  mean
```

```
##          0.0853  7e-04
## s.e.    0.0198  2e-04
##
## sigma^2 estimated as 5.978e-05:  log likelihood=8706.18
## AIC=-17406.37   AICc=-17406.36   BIC=-17388.86
```

```
auto.arima(crsp,max.q=0,approximation=T)
```

```
## Series: crsp
## ARIMA(2,0,0) with non-zero mean
##
## Coefficients:
##          ar1          ar2    mean
##          0.0865   -0.0141   7e-04
## s.e.    0.0199    0.0199   2e-04
##
## sigma^2 estimated as 5.979e-05:  log likelihood=8706.43
## AIC=-17404.87   AICc=-17404.85   BIC=-17381.53
```

Based on AIC, when we set approximation to be false, AR(1) is the best AR(p) model. However, if we set approximation to be true, the result differs. AR(2) is the best model for AR(p) when approximation is true.